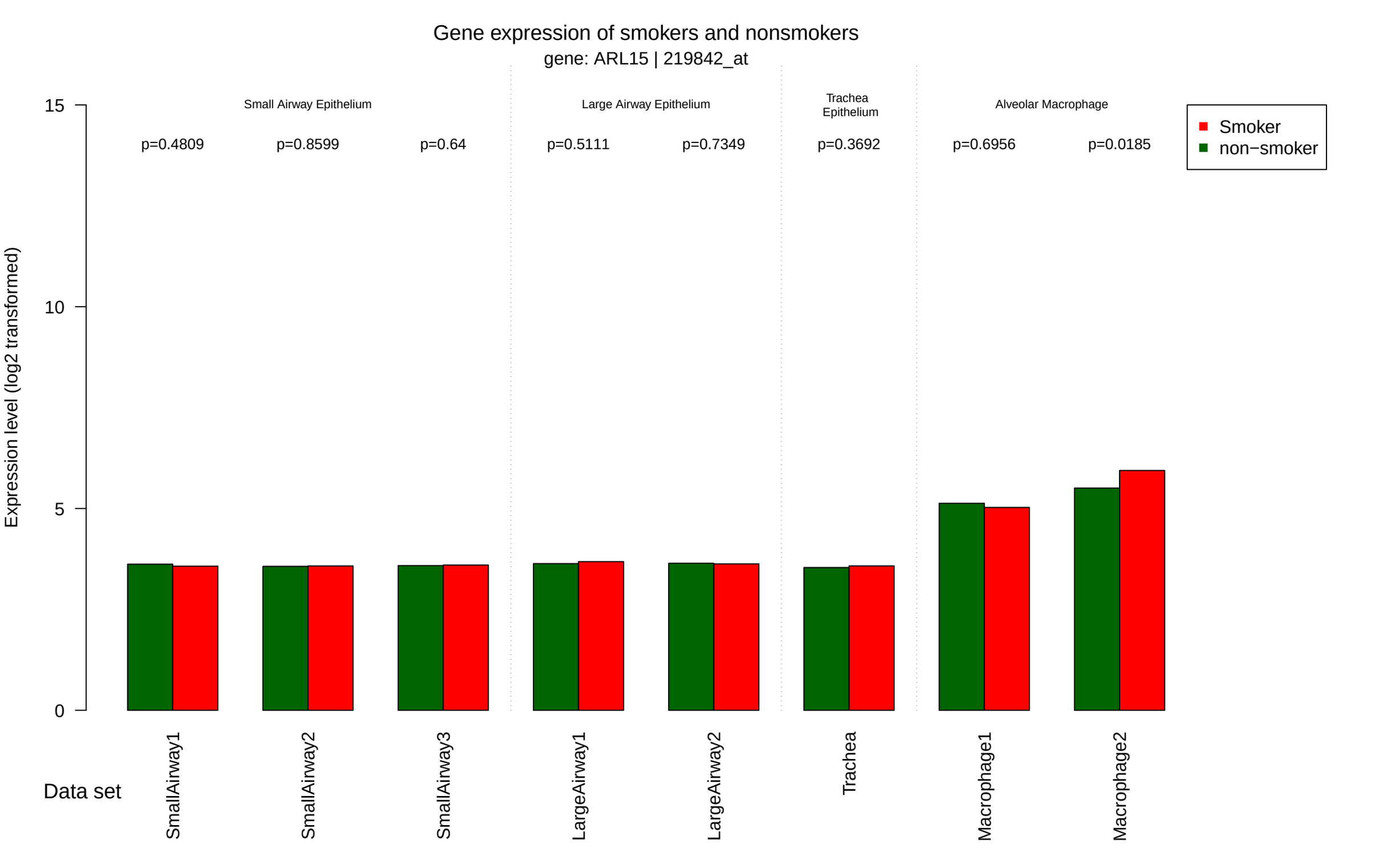
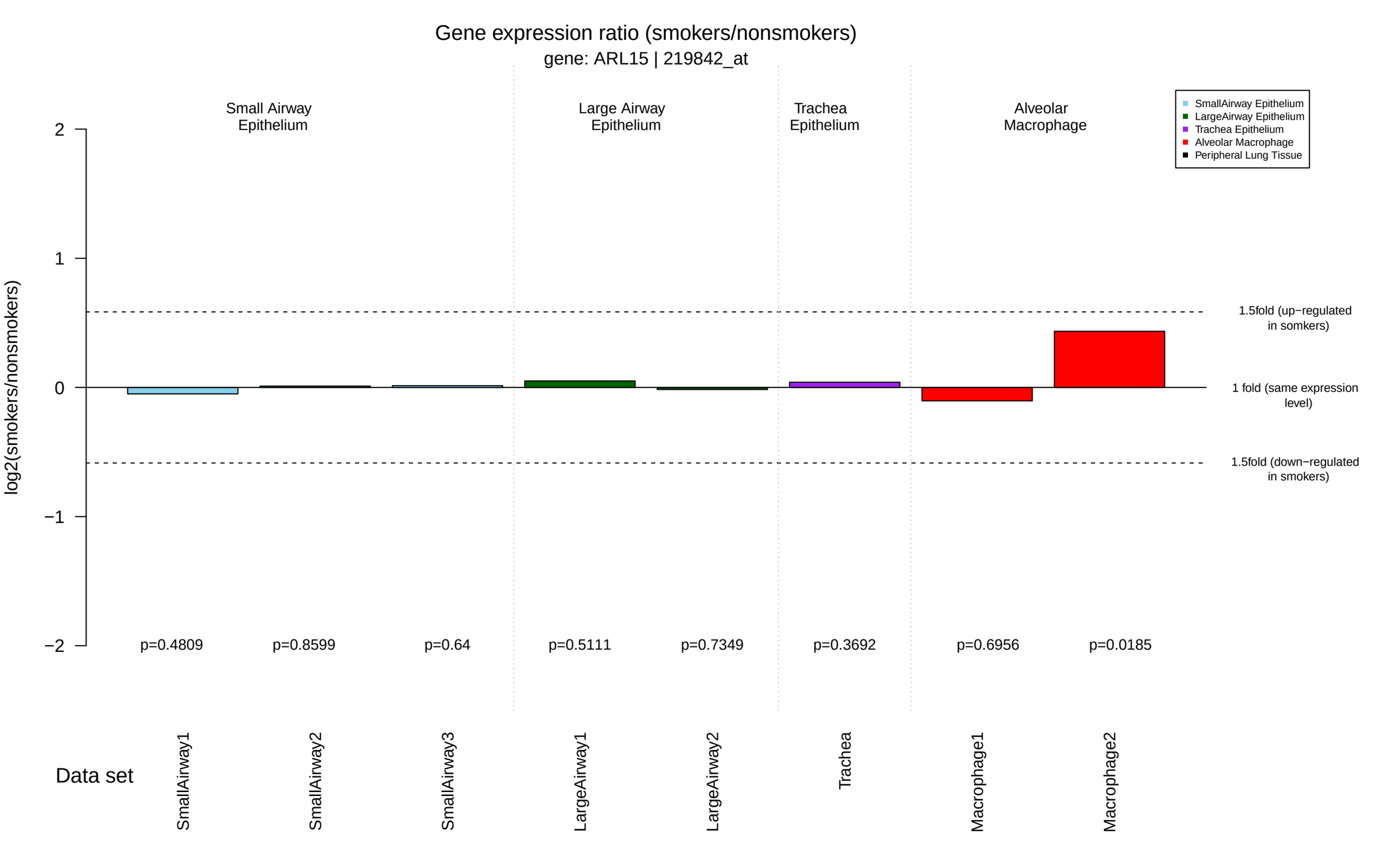
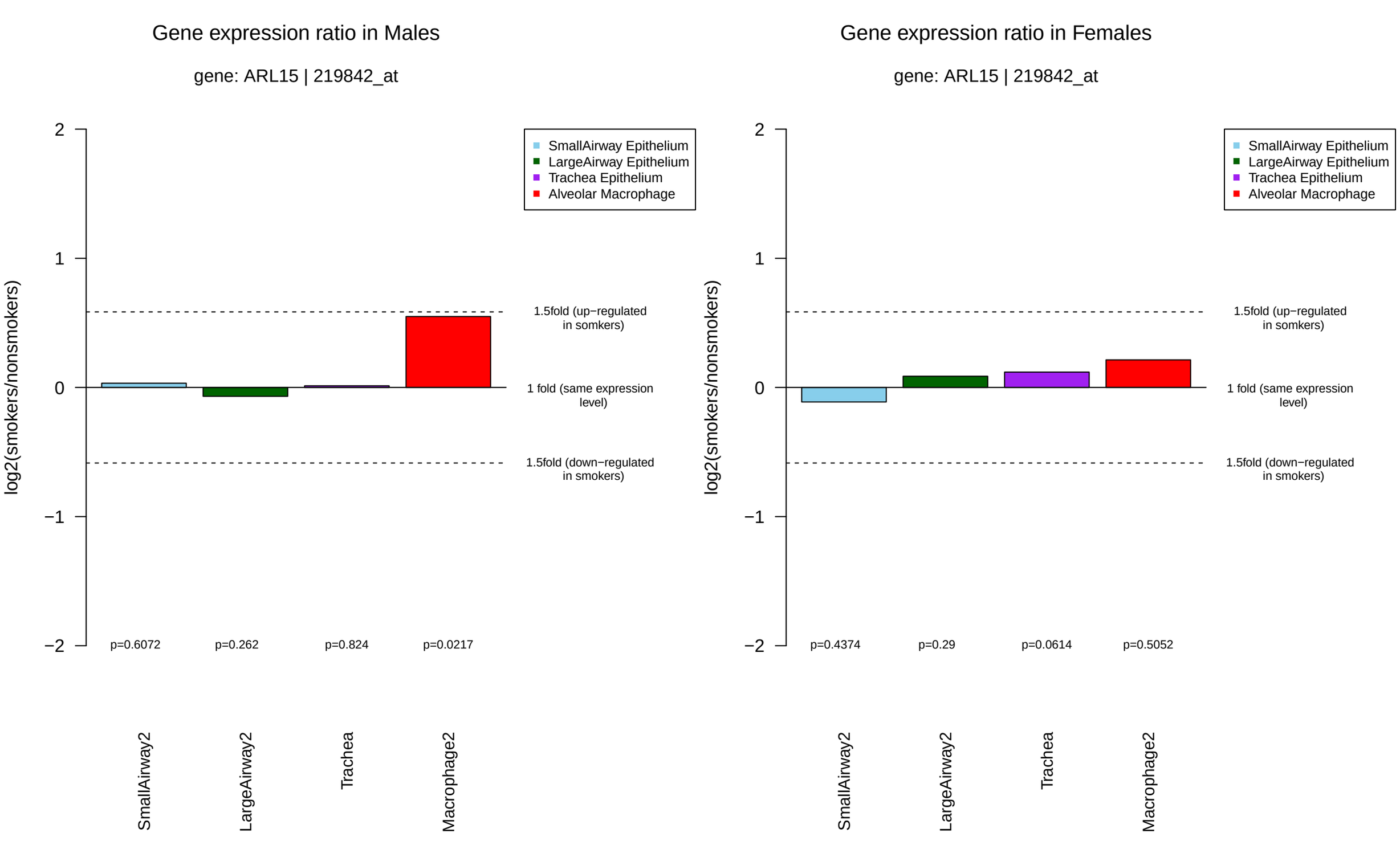
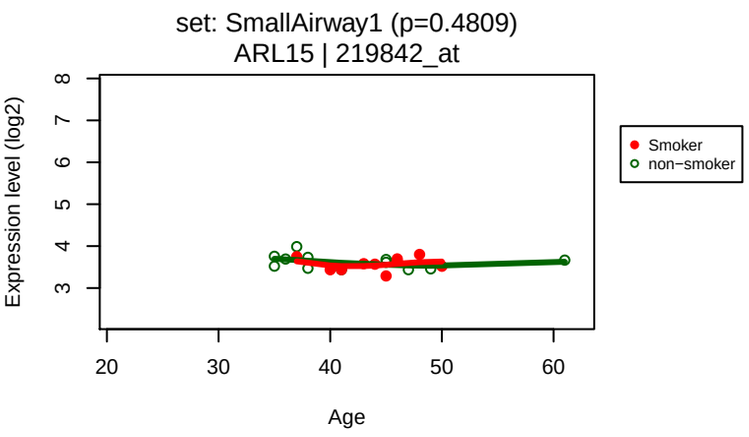
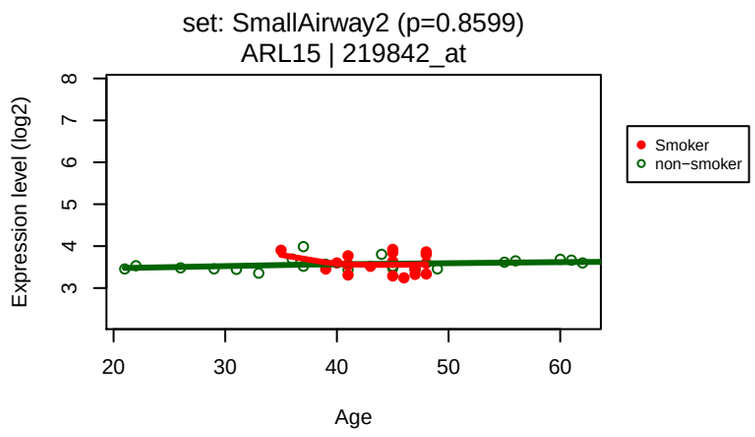
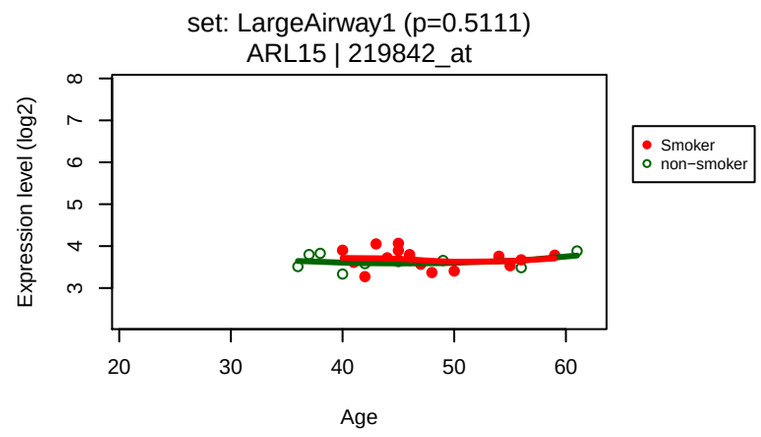

Gene expression of smokers and nonsmokers
gene: ARL15 | 219842_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4809
p=0.8599
p=0.64
p=0.5111
p=0.7349
p=0.3692
p=0.6956
p=0.0185
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

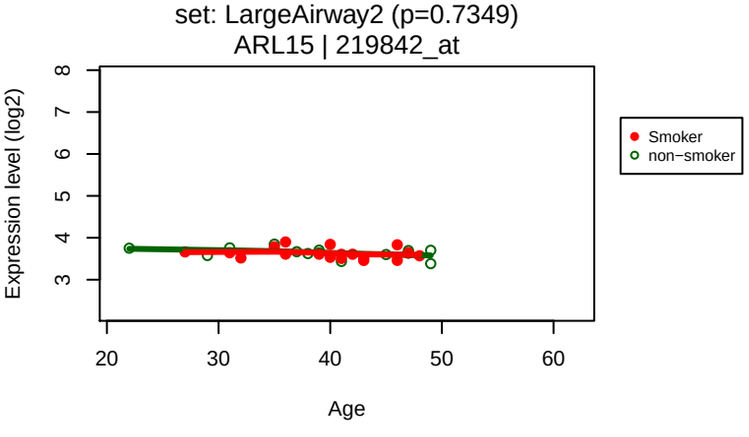

Gene expression ratio (smokers/nonsmokers)
gene: ARL15 | 219842_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4809
p=0.8599
p=0.64
p=0.5111
p=0.7349
p=0.3692
p=0.6956
p=0.0185
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

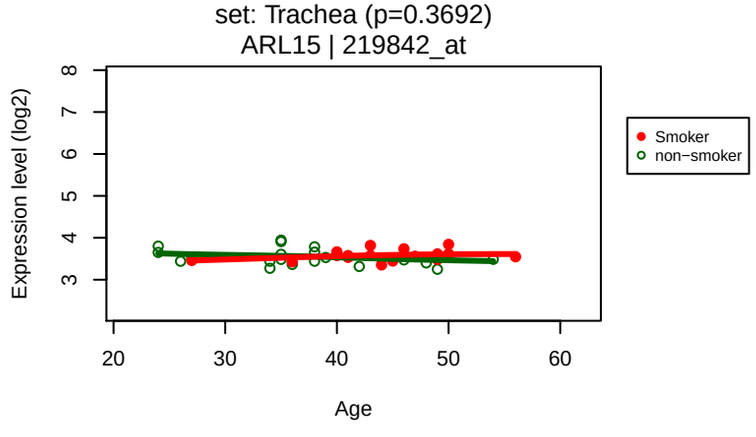

Gene expression ratio in Males
gene: ARL15 | 219842_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6072
p=0.262
p=0.824
p=0.0217
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL15 | 219842_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4374
p=0.29
p=0.0614
p=0.5052
SmallAirway2
LargeAirway2
Trachea
Macrophage2

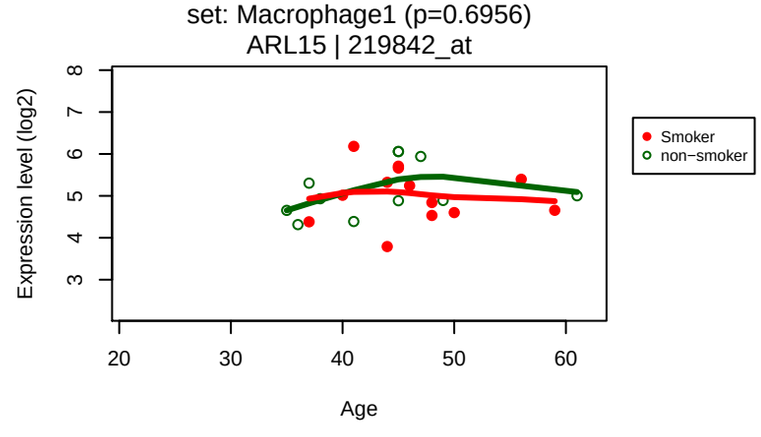

set: SmallAirway1 (p=0.4809)
ARL15 | 219842_at
Expression level (log2)
Age
Smoker
non−smoker

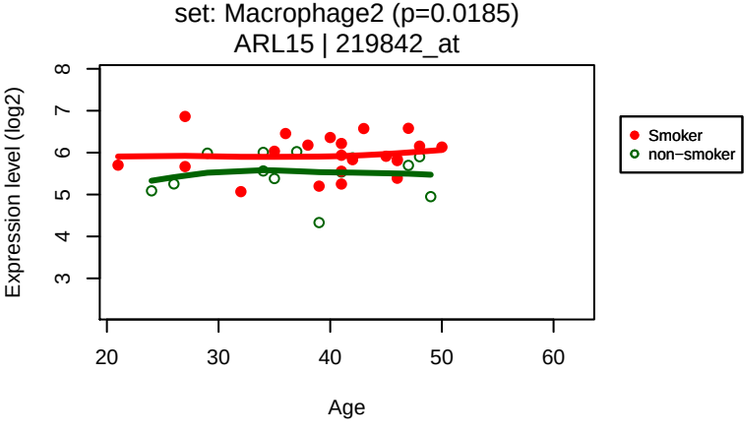

set: SmallAirway2 (p=0.8599)
ARL15 | 219842_at
Expression level (log2)
Age
Smoker
non−smoker

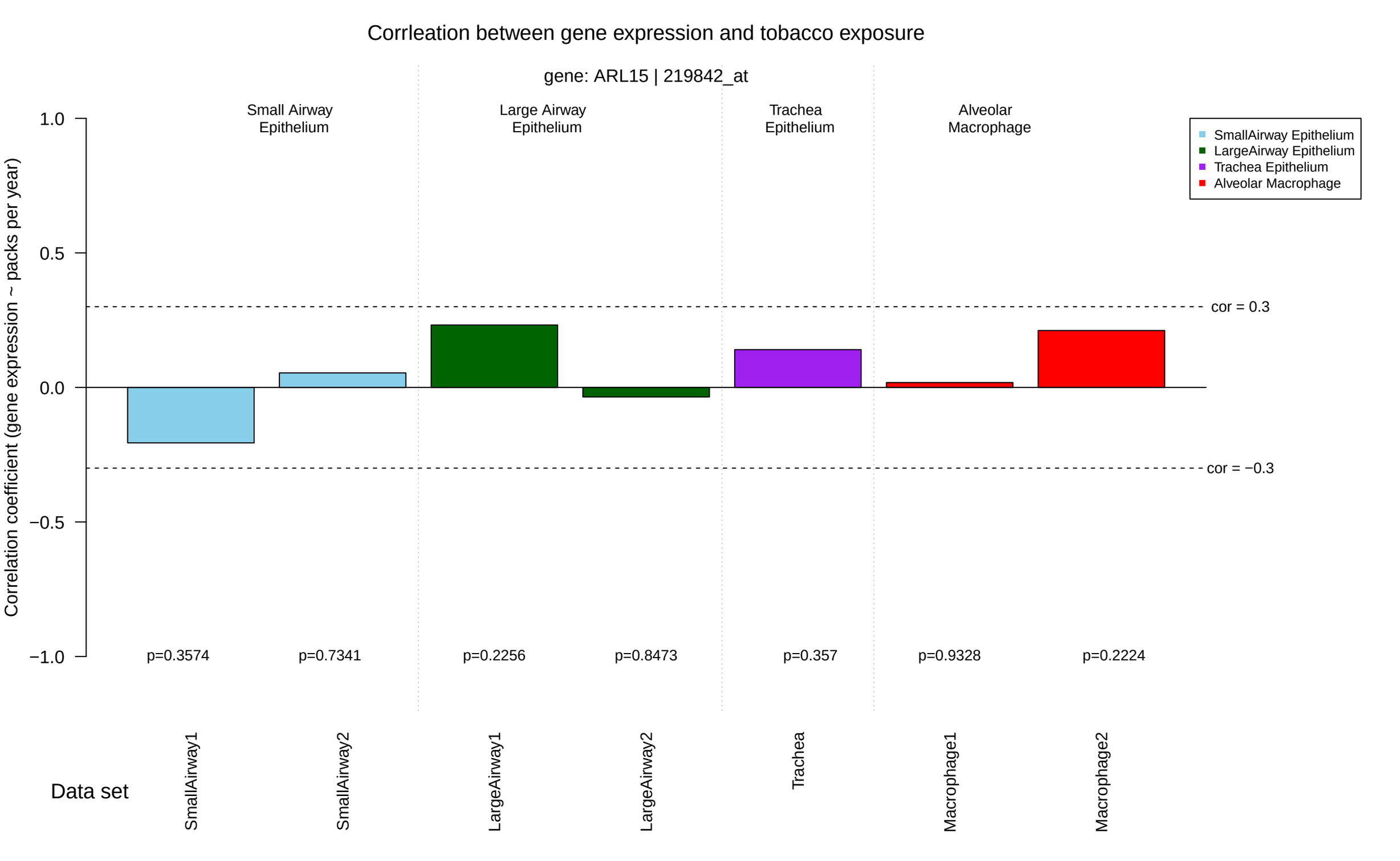

set: LargeAirway1 (p=0.5111)
ARL15 | 219842_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7349)
ARL15 | 219842_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3692)
ARL15 | 219842_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6956)
ARL15 | 219842_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0185)
ARL15 | 219842_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL15 | 219842_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3574
p=0.7341
p=0.2256
p=0.8473
p=0.357
p=0.9328
p=0.2224
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2